Biochemistry Cumulative Examination

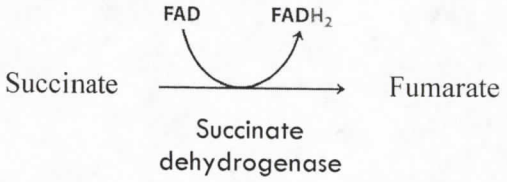
Biochemistry (NOT genetics) of branched-chain amino acids and branched chain aminotransferase.
Suggested reading for this announced cumulative exam:
Crystal structures of human mitochondrial branched chain aminotransferase reaction intermediates: Ketimine and pyridoxamine phosphate forms. *Biochemistry* 2002, **41**, 11592.
Branched-chain amino acid catabolism is a conserved regulator of physiological ageing. *Nature Comm* 2015, **6**, 10043.
New insights into the role of the branched-chain aminotransferase proteins in the human brain. *J Neurosci Res* 2015, **93**, 987.
Mechanisms of amino acid-mediated lifespan extension in Caenorhabditis elegans. *BMC Genet* 2015, **16**, doi 10.1186.
Relevant sections in your biochemistry textbook.

In your structures drawn for this exam, please use skeletal structures.

1. Draw the branched chain amino acids that branched chain aminotransferase reacts with and name each structure. There is one amino acid, which while it has a branched chain, is *not* a substrate for this enzyme according to some of the papers. Name it and clearly identify it as the one that is not a substrate. 15 points

2. Using any representative of the branched chain amino acids show, in a style similar to the example below, a reaction catalyzed by branched chain aminotransferase. You may use words or structures for the substrate(s), product(s), and cofactor(s). 15 points

EXAMPLE



3. Show a detailed chemical mechanism (*i.e.* push electrons) for the reaction you gave for question 2. 15 points

4. In addition to their main substrates, all the aminotransferases react with a particular amino acid and its precursor. Name the amino acid and its precursor and draw their structures. 10 points

5. Edwards *et al.* assert that "anaplerosis of tricarboxylic acid (TCA) cycle substrates likely plays a role in lifespan extension."
a. Define anaplerosis. 5 points
b. What evidence did they present which lead them to this conclusion? 10 points

6. Mansfeld *et al.* also mention anaplerosis, but emphasize the impact of BCAA on neurons. What evidence did they present which shows neurons are important in nematode lifespan extension by BCAA? 10 points

7. Hull *et al.* also highlight neurons, but this time in humans. Instead of BCAAs themselves, they point toward that other amino acid (question 4). Why is this other amino acid important in neurons? 10 points

8. They also discuss the CXXC motif. In three or four sentences, what role does this play in regulation of branched chain aminotransferase activity? 10 points